MULITILINGUAL TRANSLATION TEAM

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Problem Statement

Most Python learning platforms are available only in English, making it difficult for users who speak other languages. Many translation systems also translate Python code and programming keywords incorrectly, which creates confusion. In addition, using multiple AI models together in the cloud increases deployment and maintenance costs.

Objective:-

The objective of this team is to get and deliver the text,video,audio in multi languages

Solution:-

The proposed system has two modules:

• Pre-recorded Tutorial Translation – Translates tutorial videos into different languages while keeping the instructor's voice and Python code unchanged.

• Real-Time User Translation – Allows users to ask questions using voice or text and receive translated responses instantly. The system detects and protects Python code before translation and restores it after translation to avoid syntax errors.

Technologies Used

Component	Technology / Model
Programming Language	Python
Backend	FastAPI
Speech-to-Text	Whisper Large-v3, Parakeet TDT 0.6B
Translation	IndicTrans2, NLLB-200
Language Detection	fastText
Voice Detection	Silero VAD
Voice Output	OpenVoice V2 / XTTS-v2 / CosyVoice 2
API	REST API

Cost Optimization

To reduce cloud cost, each AI model is deployed as a separate FastAPI service instead of loading all models into one application. This reduces GPU usage, improves resource management, and allows each service to run independently. The original architecture and AI models remain unchanged.

Expected outcome

The system will help users learn Python more easily in their native language.